QP Code: _____ Reg. No.: .....
Name : .....…

MAHATMA GANDHI UNIVERSITY, KOTTAYAM MGU-UGP (HONOURS) EXAMINATION

SEMESTE;R 5 (MODEL QUESTION PAPER)

MG5SECPHY301: Physics using Computational Tools

Time: 1.5 Hours **Max. Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

PART A (Answer any 10 questions)

1. Using the Newton-Raphson method, find the first approximation x_1 to a root of $f(x) = x^2 - 2$, starting from $x_0 = 1$. *[CO1]*

2. Using the Regula-Falsi method, find the first approximation to a root of $f(x) = x^2 - 4$ in the interval [1, 3]. *[CO1]*

3. For the system $x + y = 5$, $2x + 3y = 13$, eliminate x from the second equation using the Gauss elimination procedure and write the resulting equation. *[CO1]*

4. Write the coefficient matrix A and the constant vector B for the system $2x + y = 5$, $x + 3y = 10$. *[CO2]*

5. For the data points (1, 2) and (3, 8), find the slope b of the least-squares straight line $y = a + bx$ joining them. *[CO1 & CO2]*

6. Given $f(0) = 1$ and $f(1) = 3$, find the forward difference $\Delta f(0)$. *[CO1 & CO2]*

7. Given the points (0, 1), (1, 3) and (2, 7), find the divided difference f[0, 1]. *[CO1 & CO2]*

8. Given $f_0 = 1$ and $\Delta f_0 = 2$ at $x_0 = 0$ with $h = 1$, estimate $f(0.5)$ using the linear term of Newton's forward difference formula. *[CO1 & CO2]*

9. Given $f(1) = 2$ and $f(3) = 8$, estimate $f'(1)$ using the two-point approximation $f'(1) \approx (f(3) - f(1)) / (3 - 1)$. *[CO3 & CO4]*

10. Using the trapezoidal rule, find the area under the curve between $x = 0$ and $x = 1$, given $f(0) = 2$ and $f(1) = 4$. *[CO3 & CO4]*

11. For $dy/dx = x + y$ with $y(0) = 1$, calculate the value of dy/dx at $x = 0$. *[CO3 & CO4]*

12. In Euler's method, given $y_0 = 1$, $f(x_0, y_0) = 2$ and $h = 0.1$, find y_1. *[CO3 & CO4]*

[2 X 10 = 20]

PART B (Answer any 5 questions)

13. Using the Newton-Raphson method, find the root of the equation $x^2 - 5 = 0$, correct to three decimal places, starting with $x_0 = 2$ (perform two iterations). *[CO1]*

14. Solve the following system of equations using the Gauss elimination method: *[CO1]*

$$x + y + z = 6$$
$$2x - y + z = 3$$
$$x + 2y - z = 2$$

15. Fit a straight line $y = a + bx$ to the following data using the method of least squares. *[CO1 & CO2]*

x	1	2	3	4
y	3	5	7	8

16. Using Newton's forward difference interpolation formula, estimate the value of f(2.5) from the following data. *[CO1 & CO2]*

x	1	2	3	4
f(x)	2	5	10	17

17. Using Lagrange's interpolation formula, find the value of y at x = 2 for the following data. *[CO1 & CO2]*

x	0	1	3
y	1	3	13

18. Using the trapezoidal rule, evaluate the integral of f(x) between x = 0 and x = 4 from the following table (n = 4 intervals). *[CO3 & CO4]*

x	0	1	2	3	4
f(x)	1	2	5	10	17

19. Using the 4th order Runge-Kutta method, solve $dy/dx = x + y$, $y(0) = 1$, and find $y(0.1)$ taking a single step of size $h = 0.1$ (show all four k values). *[CO3 & CO4]*

20. Using Euler's method, solve $dy/dx = x + y$, $y(0) = 1$, and find $y(0.2)$ taking step size $h = 0.1$ (perform two steps). *[CO3 & CO4]*

[5 × 6 = 30]